

Worksheet 2

For questions 1 through 4, draw three indifference curves for each of the following descriptions. Provide a separate graph for each answer. Label your axes and the indifference curves (e.g. IC_1 , IC_2 , IC_3). When applicable, identify specific coordinates of points on the indifference curves. Indicate with an arrow which direction is “better” as done in class. This worksheet is out of 20 points.

1. Michael likes to spend time with his girlfriend Holly but dislikes spending time with his HR manager Toby. q_1 = time spent with Holly (minutes) and q_2 = time spent with Toby (minutes).
2. Kevin always wants to make his chili with beef and onions at a ratio of 2 lbs beef to 1 lb onions. More chili is preferred to less, but extra ingredients beyond this ratio do not increase or decrease his utility. q_1 = beef (lbs) and q_2 = onions (lbs). Provide the coordinates of the relevant corner points for each IC.
3. Angela loves cats of all colors. She only cares about the total number of cats she owns, regardless of their coloring. q_1 = black cats and q_2 = orange cats. Provide coordinates of the intercepts of the ICs drawn.
4. Nard Dog likes to play guitar but is indifferent to spending time with his brother. Meaning, he finds bundles where more time or less time with his brother equally preferred as long as he gets to play guitar the same amount. Let q_1 = time spent playing guitar and q_2 = time spent with his brother.
5. Frank's utility function is $U(q_1, q_2) = 3q_1 + 6q_2$.
 - (i) Compute the marginal utilities.
 - (ii) Compute the marginal rate of substitution (MRS).
 - (iii) On a graph, draw indifference curves for $U = 60$ and $U = 120$, labeling the horizontal and vertical intercept coordinates.
6. Dee's utility function is $U(q_1, q_2) = \min\{2q_1, q_2\}$. On a graph, draw indifference curves for $U = 10$ and $U = 20$, labeling the coordinates of the corner point.
7. Charlie has the utility function $U(q_1, q_2) = q_1^2 q_2$.
 - (i) Compute the marginal utilities.
 - (ii) Compute the marginal rate of substitution (MRS). Be sure to simplify the equation where possible.
8. Dennis has the utility function $U(q_1, q_2) = q_1^{1/2} q_2^{3/2}$.
 - (i) Compute the marginal utilities.
 - (ii) Compute the marginal rate of substitution (MRS). Be sure to simplify the equation where possible.
9. Lewis from La Jolla consumes food and housing. He has a utility function of $U(F, H) = F^{1/3} H^{2/3}$, where F is the quantity of food and H is the quantity of housing. Treat F as the horizontal axis good. Find Lewis's MRS.